

PADMAPRIYA RAJASEKARAN

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SUMMARY

Biomedical Engineer with 1.5 years of industry experience working on FDA-regulated biomechanical devices

EDUCATION

Rochester Institute of Technology

Bachelor of Science in Biomedical Engineering, Chemistry Minor

Aug 2021 - Present

Dean's List

SKILLS

Engineering: Verification & Validation, Root Cause Analysis, IQ/OQ/PQ, DFMEA, Gage R&R, Functional & Regression Testing, NPI, Creating Work Instructions & SOPs, Deviation & Non-Conformance Management, SCADA

Software: MATLAB, Fusion 360, FEA, Creating Work Instructions & SOPs, JIRA, Minitab.

Certifications: Lean Six Sigma Green Belt (C.S.S.C)

WORK EXPERIENCE

Fresenius Medical Care - NPI Systems Engineering Co-op

Jul 2025 - Dec 2025

- Developed 8D report, performing root cause analysis and implemented corrective actions on Home Dialysis verification test setup reducing the measurement error to 0.6%
- Established process specification for 7 different process parameters and set up control limits with triggers for continuous monitoring
- Developed a SAT protocol for newly installed reliability test station and conducted MSA validation
- Developed a MATLAB-based data pipeline to gather real-time test data to generate machine health reports
- Performed fault injection testing to verify systems safe operation limits and documented the findings in equipment specification record
- Conducted DOE trials with multiple parameter combinations to validate sensor performance
- Documented process deviations using Good Documentation Practices and presented findings to cross-functional teams for timely disposition
- Tested and approved unreleased hardware for backward compatibility, with three different industry-standard hardware attachments
- Improved and replaced the existing test procedure by standardizing hardware and by adding and optimizing sensor location and increased the confidence level beyond 95%
- Performed Monte Carlo simulations on test data to establish optimized manufacturing process parameters

Cranial Technologies - R&D Engineering Co-op

Jul 2023 - May 2024

- Worked with Cross functional organization to change manufacturing process of infant cranial helmets from CNC milling to 3D printing and reduced the cost by 17%
- Developed and Validated Quality inspection SOPs for newly launched product lines
- Worked on generating visual inspection checklist for 15 defects across four different product lines
- Automated 3D scan data conversion by developing a MATLAB tool to process raw 3D scan data into 3D printable STL data, reducing preprocessing time
- Optimized 3D printer build volume, increasing throughput by 53% while maximizing material utilization
- Performed destructive testing to analyze mechanical and functional durability of component and bench marked data used to demonstrate substantial equivalence in the device's FDA submission
- Commissioned an HP Multi Jet Fusion 3D printer for production, collaborating with HP engineers to troubleshoot and hand off for full floor integration
- Collaborated with cross functional team to author 9 WI and trained 3 operators for 3D printing and Vapor deposition operations
- Designed and built custom jigs, fixtures using GD&T principles to ensure dimensional accuracy of 3 part assembly

PROJECTS

Contact Lens Dry Processing: Redesigned the workflow to integrate two process variables, boosting yield by 19% and eliminating stand-down time

Intravenous Glucose Monitoring System: Benchmarked sensor costs (\$30–\$350) across 16 different options and contributed to the decision matrix. Redefined problem statement for continuous glucose monitoring. Developed an FMEA for the continuous glucose monitoring device

Button Controlled Key Turner: Designed and manufactured one handed key turning device and down selected raw material based on safety factor from FEA analysis. Defined must-have design criteria and conducted stakeholder analysis across OTs, patients, insurers, and regulators, translating clinical and market needs into a testable device specification

Air Filtration System: Designed a filtration system to reduce pollutants and lower noise levels to 62.8 dB

Physiotherapy Robot: Designed a 6-axis therapy robot as a low cost alternative to in-person physical therapy tutor